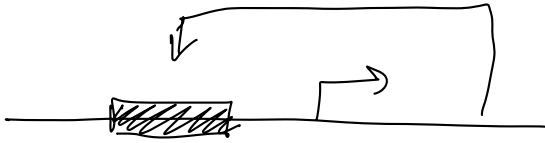


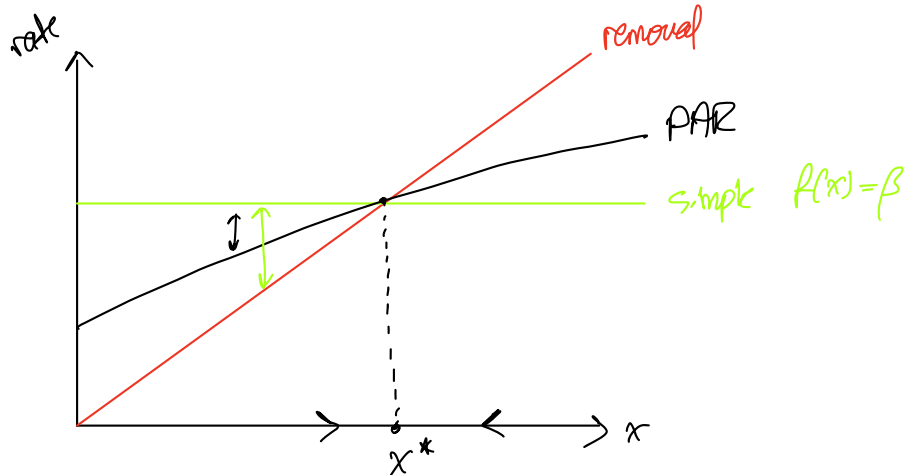
Positive Autoregulation

Developmental networks $\rightarrow$ irreversible changes.

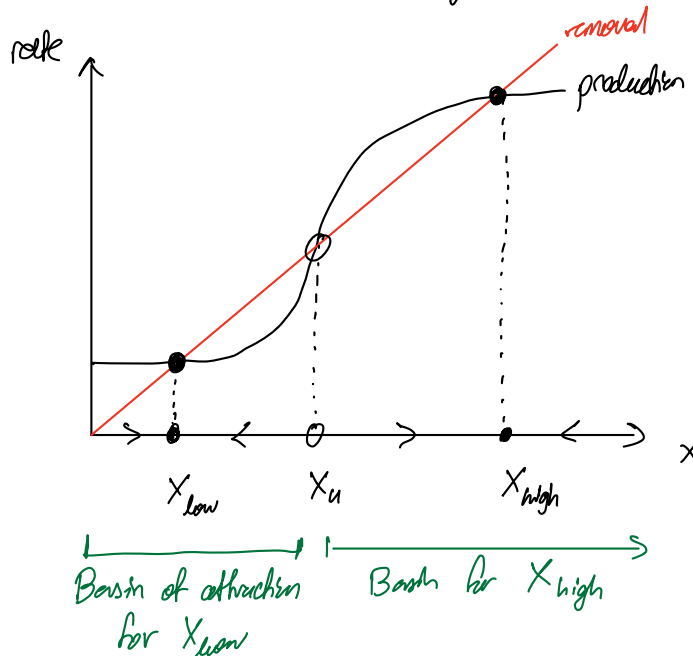
A common motif is Positive Autoregulation:



Dynamics: $\dot{x} = f(x) - \alpha x$



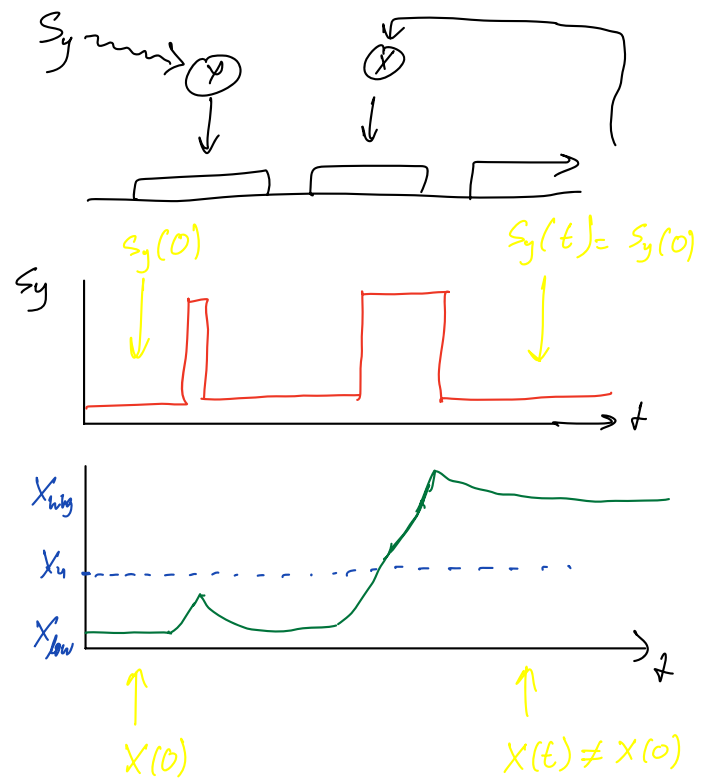
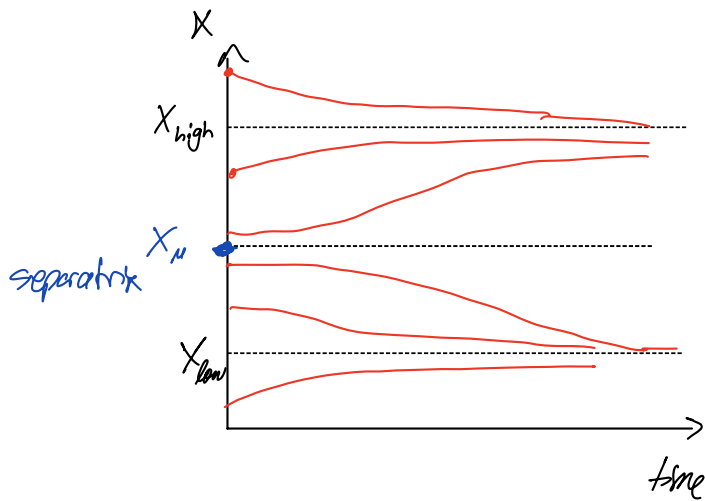
If $f(x)$ is cooperative, we can get Bistability:



(let class do the arrows & stability)

The system thus has MEMORY of initial conditions: start in a basin, end in a basin.

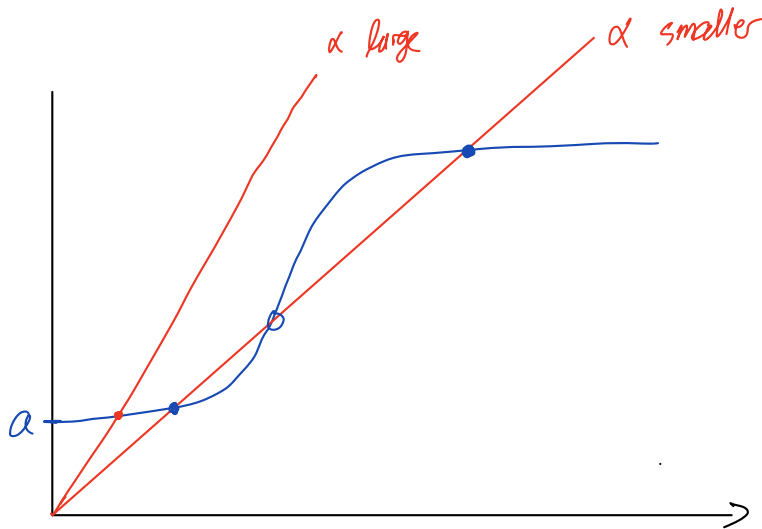
(draw point, ask where it will go).



Bistability is not a guaranteed outcome: Consider:

$$f(x) = \beta + \frac{\beta_1 (x/\kappa)^n}{1 + (x/\kappa)^n}$$

$$\dot{x} = f(x) - \alpha x$$



Here, α is the control parameter for a bifurcation.

↓
change in the # or stability of F.P.s

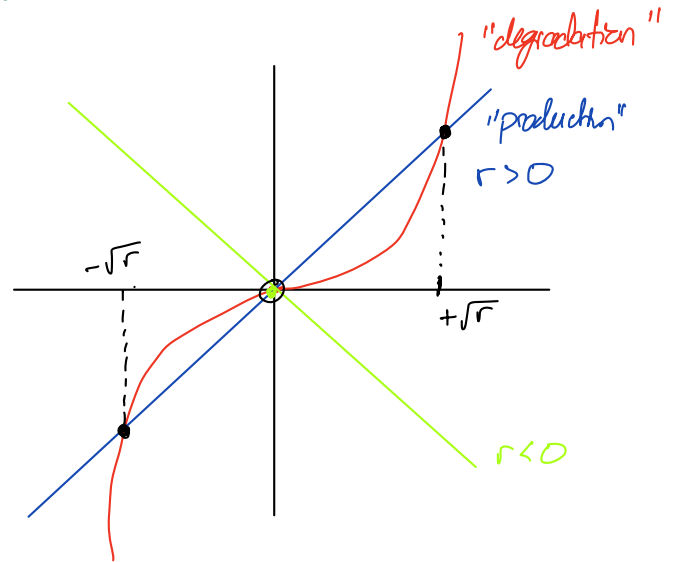
Explicitly solvable example of bistability

Let us consider the dynamics

$$\dot{x} = r x - x^3 = 0$$

either $x=0$, or $x = \pm\sqrt{r}$

if $r < 0$, only $x=0$.



Stability

Suppose I couldn't draw - how would I determine stability?

In general: $\dot{x} = g(x)$

Consider a fixed point x^* satisfying $g(x^*) = 0$.

Consider a small perturbation: $x(t) = x^* + \delta x(t)$

$$\begin{aligned} \frac{d}{dt} \delta x &= g(x^* + \delta x) = g(x^*) + g'(x^*) \delta x + \frac{g''(x^*)}{2} \delta x^2 + \dots \\ &\simeq g'(x^*) \delta x \end{aligned}$$

Stability condition:

$$g'(x^*) < 0$$

$$g'(x) \equiv J(x) \text{ Jacobian}$$

Example: $g(x) = rx - x^3 \Rightarrow g'(x) = r - 3x^2$

$$J(0) = r, \quad J(\pm\sqrt{r}) = -2r$$

Energy

Notice:

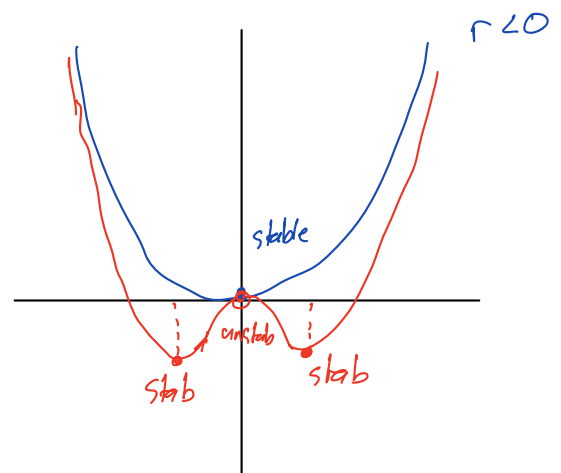
$$\dot{x} = -\frac{\partial U}{\partial x}, \quad \text{where} \quad U(x) = -\frac{r}{2} x^2 + \frac{x^4}{4}$$

Plot for $r > 0$ or $r < 0$:

$x(t)$ is like a ball rolling down a hill.

It settles when $U'(x) = 0 \Rightarrow$ at the minima of U .

Maxima correspond to unstable fixed points.



If I start at $x=0$ and kick it, where does it go? for $r < 0$?
for $r > 0$?

This "energy" picture always works in one dimension:

$$\dot{x} = f(x) - \alpha x = -\frac{\partial U}{\partial x}, \quad U(x) \equiv \frac{\alpha x^2}{2} - \int dx f(x).$$

Two-node feedback loops

Double-positive loop: $Y \rightleftarrows X$

Either both on, or both off.

Double-negative loop: $X \rightleftarrows X$

One on, other off $\rightarrow$ called a toggle switch.

These also display bistability:

$$\dot{x} = f(y) - \alpha x$$

$$\dot{y} = f(x) - \alpha y$$

It is helpful to draw a phase portrait

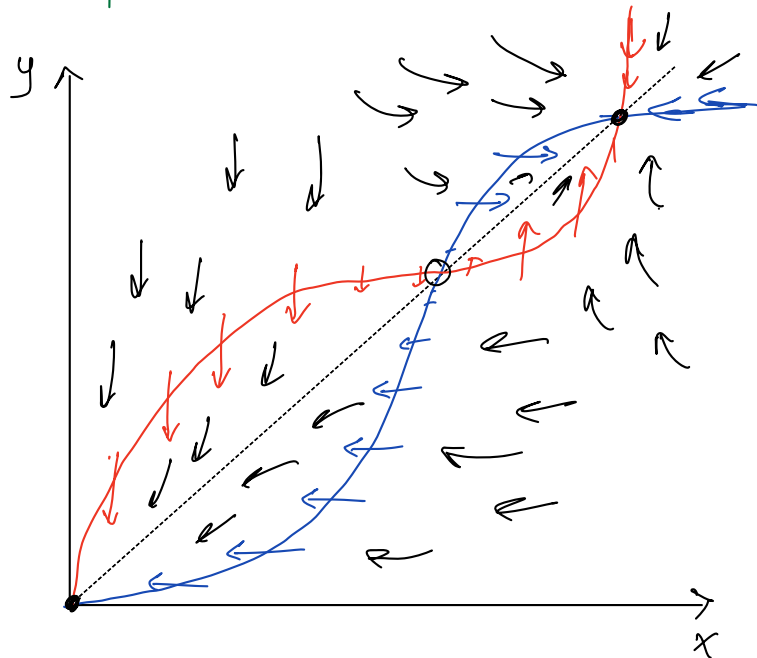
Idea: draw the x - y plane, indicate fixed points and nullclines, and indicate the flow.

Nullcline: lines where one of the variables is not changing.

Above: $\dot{x} = 0 \Rightarrow \boxed{x = f(y)/\alpha}$

$\dot{y} = 0 \Rightarrow \boxed{y = f(x)/\alpha}$

Fixed points: $\dot{x} = \dot{y} = 0 \Rightarrow$ intersection of nullclines



Nondimensionalization

When we say a dimensional parameter is "large" or "small", the statement only makes sense when compared against some other dimensional parameter.

To reveal characteristic scales, it is often useful to nondimensionalize:

$$\frac{dX}{dt} = \beta \frac{(X/K)^n}{1 + (X/K)^n} - \alpha X$$

Dimensions? 3 params.

Define $x \equiv X/K$, $\tau = \alpha t$ ← units of dilution time (dimensionless).

Use $\frac{d}{dt} = \frac{d\tau}{dt} \frac{d}{d\tau} = \alpha \frac{d}{d\tau}$

$$\alpha K \frac{dx}{d\tau} = \beta \frac{x^n}{1 + x^n} - \alpha K x$$

$$\frac{dx}{d\tau} = \bar{\beta} \frac{x^n}{1 + x^n} - x$$

$$\bar{\beta} \equiv \frac{\beta}{\alpha K} \Rightarrow \text{There is only one independent control parameter in the problem}$$

Comment on Networks